



Physical & Functional Understanding: From “What It Is” to “What It Can Do”

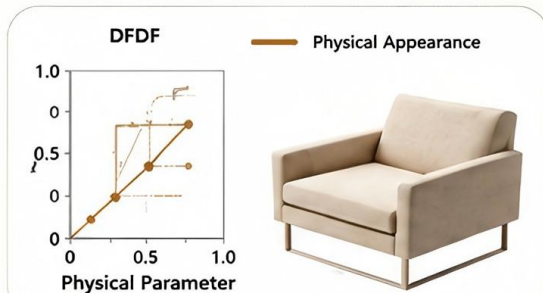
Bridging visual perception, physical reasoning and embodied interaction

01



Physical Property Estimation

Infer material and mechanical properties of objects, including appearance, mass, rigidity and deformability, from visual observations.



02



Affordance Reasoning

Reason actionable possibilities of objects, evolving from static geometry to human demonstration and active interactive learning.

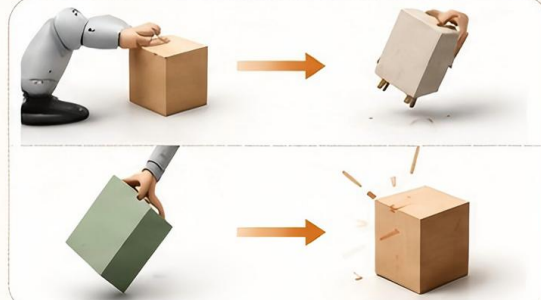


03



Physical Causal Reasoning

Predict actions outcome, temporal dynamics and counterfactual physical events.

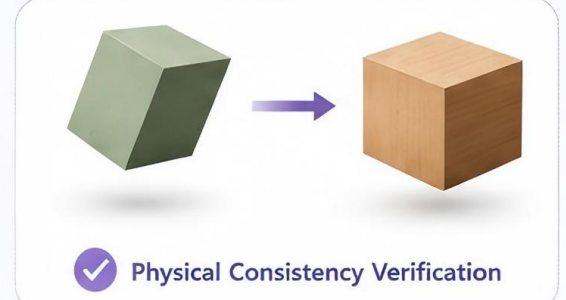


04



Physical Consistency Verification

Evaluate physical plausibility and validate executable simulation assets beyond visual realism.



Perceive the World



Understand Properties



Infer Affordances



Predict Consequences



Act in the Physical World